

Haskell [1935] revisited

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Abstract. The *Haskell* [1935] value of 10^{21} Pa s for mantle viscosity is a classic and enduring constraint on the rheology of the Earth's interior. We revisit this inference using spherically symmetric, self-gravitating, Maxwell viscoelastic Earth models. Our inference is based on both forward and inverse analyses of decay times associated with uplift at two sites considered by Haskell, Angerman River, Sweden, and Oslo, Norway, rather than the raw relative sea level (RSL) data at these sites. We demonstrate that predictions of the decay time associated with the Angerman River data are insensitive to variations in both the late Pleistocene ice load history and the lithospheric thickness of the Earth model (the predictions at Oslo are sensitive to both these inputs), and hence decay times at this site provide a remarkably robust constraint on mantle viscosity. We derive a constraint on the "average" viscosity of the mantle of $0.65 - 1.10 \times 10^{21}$ Pa s, where the "average" resolved by the data encompasses a region which extends from the base of the lithosphere to a depth of near 1400 km. This indicates that many previous analyses which have invoked the Haskell value of 10^{21} Pa s as a constraint on the average upper mantle (i.e., above 670 km depth) viscosity alone have misinterpreted the resolving power of the inference. Furthermore, our analysis indicates that a number of apparently contradictory inferences of viscosity based on Fennoscandian data satisfy the new, rigorous, interpretation of the Haskell constraint. Finally, we demonstrate how the ambiguity in the upper mantle/lower mantle viscosity contrast associated with the Haskell "average" may be reduced by invoking decay time constraints estimated from RSL curves in Hudson Bay. A preliminary inversion of decay times at Angerman River and Richmond Gulf (in Hudson Bay) suggests a contrast of approximately an order of magnitude between the average viscosities of the upper mantle and the top 1000 km of the lower mantle; however, a conclusive analysis in this regard must await the determination of consistent decay time estimates for the Hudson Bay region.

Introduction

Inferring mantle viscosity from the analysis of data associated with the glacial isostatic adjustment (GIA) of the planet is a classic problem in geophysics. Unfortunately, in spite of a large body of published work spanning at least 60 years, little, if any, consensus has emerged. Indeed, recent inferences of viscosity are discrepant by an order of magnitude or more at all depths [e.g., *Nakada and Lambeck*, 1989; *Fjeldskaar and Cathles*, 1991; *Tushingham and Peltier*, 1992; *Han and Wahr*, 1995].

The sources of this disagreement are varied. First, numerical predictions of GIA require as input not only a viscoelastic Earth model but also the space-time history of the late Pleistocene ice sheets. The latter (in addition to the former) is imprecisely known, and in

recent years there has been a concerted effort to seek parameterizations of the GIA database which are insensitive to this uncertainty. *Nakada and Lambeck* [1989], for example, have considered differential Late Holocene highstands at various pairs of far-field (primarily Australian) sites, while *Mitrovica and Peltier* [1993a, 1995] have argued that the characteristic decay times of post-glacial relative sea level (RSL) curves obtained near the center of previously glaciated regions provide a relatively "uncontaminated" measure of mantle viscosity.

The discrepancy associated with previous viscosity inferences may also be influenced by the presence of lateral heterogeneities in rheology [e.g., *Sabadini et al.*, 1986; *Gasparini and Sabadini*, 1989; *Manga and O'Connell*, 1995]. This possibility is particularly problematic for models of the radial profile of mantle viscosity based on globally distributed data sets [e.g., *Tushingham and Peltier*, 1992]; however, it does not explain apparently contradictory inferences based on data from the same region (e.g., Fennoscandia [*McConnell*, 1968; *Lambeck et al.*, 1990; *Fjeldskaar and Cathles*, 1991]).

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Finally, limitations in the inference procedure may also play an important role in the debate. The vast majority of previous viscosity estimates have been based on a limited suite of forward calculations. As a consequence, the resolving power of the GIA data has generally not been determined nor have uncertainties in the viscosity estimates been established. To overcome these limitations, *Mitrovica and Peltier* [1991a] have advocated rigorous inversion procedures. The application of such approaches to the Hudson Bay RSL record [*Mitrovica and Peltier*, 1995] has yielded a bound on the average viscosity between about 400 km and 1800 km depth of $1 - 4 \times 10^{21}$ Pa s, with the upper bound being appropriate in the southern portion of the bay.

In the present analysis we apply these new considerations to revisit perhaps the most influential, and certainly the most enduring, inference of mantle viscosity; that of *Haskell* [1935]. *Haskell* [1935] applied a theoretical formulation, based on the loading of an isoviscous, incompressible, half-space, to the postglacial uplift and rate of uplift determined for Angerman River, Sweden, located very near the inferred center of the Fennoscandian ice complex, and Oslo, Norway. Oslo is located about 500 km from Angerman River but is still within the perimeter of the Fennoscandian ice complex at last glacial maximum. *Haskell* obtained an estimate for the kinematic viscosity of 2.9×10^{17} m²/s, which refers to a dynamic viscosity of very near 10^{21} Pa s. The latter is the well-known “Haskell value.”

In the modern context, a number of important questions remain unanswered in regard to the Haskell analysis. First, is the *Haskell* [1935] inference, as he argued, insensitive to the details of the Fennoscandian melting event? Second, is it possible to extract a rigorous estimate of mantle viscosity from the Angerman River and Oslo RSL data sets, and, if so, does this estimate agree with Haskell's? Most importantly, what is the resolving power of the data? Haskell's (isoviscous) theoretical/numerical formulation obviously yields an average value of the mantle viscosity, but what is the detailed depth dependence of the averaging kernel? In particular, over what radial region does this average extend: the entire mantle, the upper mantle above 670 km depth, or some other depth range?

The lack of any detailed assessment of the latter issue lies at the heart of a fascinating, but often confusing, evolution in the interpretation of the Haskell value. *Haskell* [1936, p. 56] considered the issue and argued, in support of an earlier suggestion by *Daly* [1925], that “the subsidence of a load on the Earth's surface having a span comparable to that of a continental ice sheet must produce appreciable flow at great depth.” Specifically, *Haskell* [1936] found that at a depth equal to the width of the load the vertical displacement at equilibrium would fall to approximately 50% of the surface value. This result was based on a load in the form of an infinitely long strip; however, *Haskell* [1937] speculated that this depth would likely be reduced by less than 20% in the case of a circular load.

By the 1960s the interpretation of the resolving power of the Haskell value had been refined by a series of more

sophisticated analyses. In particular, *McKenzie* [1967, p. 297] argued that “the depth to which flow penetrates in a homogeneous half space is about equal to the radius of the applied load, or about 1000 km for Fennoscandia.” Other contemporaneous analyses clearly interpreted the Haskell number as an average value for the viscosity over roughly this range [e.g., *O'Connell*, 1971]. Nevertheless, the detailed depth dependence of the averaging kernel remained unknown.

In relatively early papers on mantle viscosity [e.g., *Anderson*, 1966; *McKenzie*, 1966, 1967; *Anderson and O'Connell*, 1967; *O'Connell*, 1971] the terms “upper mantle” and “lower mantle” were used to denote, in some general sense, the top and bottom of the mantle, and were not associated with the more modern interpretation defined by the 670-km seismic discontinuity. In fact, the concept of an upper mantle in these studies was clearly influenced by the (then) accepted resolving power of the *Haskell* [1935] value, and this region was therefore commonly associated with the top 1000–1200 km of the mantle. Furthermore, when these studies invoked degree 2 constraints (e.g., rotational anomalies) to estimate the lower mantle viscosity they invariably associated this region with the mantle below 1000–1200 km depth (or, perhaps more accurately, the region below the radial sampling range of the Fennoscandian deformation).

This (down to ~ 1200 km) interpretation of the Haskell value was supported (and in fact partly motivated) by the work of *McConnell* [1968], who derived and analyzed the so-called Fennoscandian relaxation spectrum. Using a viscous half-space model which included an elastic lithosphere and a depth-dependent viscosity, *McConnell* [1968] inferred the following features: an elastic lithosphere of thickness 120 km (a flexural rigidity 2.5×10^{25} N m); and viscosities of 4.1×10^{20} Pa s between 120 and 220 km depth, 2.7×10^{20} Pa s between 220 and 400 km depth, 10^{21} Pa s between 400 and 800 km depth, 2×10^{21} Pa s between 800 and 1200 km depth, and 6.85×10^{21} Pa s below 1200 km depth. *McConnell* invoked independent, and since invalidated [*Goldreich and Toomre*, 1969; *O'Connell*, 1971], arguments based on the nonhydrostatic bulge of Earth to suggest a several order of magnitude increase in viscosity below 1500 km depth. Although *McConnell*'s calculations were based on an entirely different data set than that used by *Haskell* [1935] (in particular, *McConnell*'s spectrum was based on data extending from the center to the maximum perimeter of the ancient Fennoscandian ice complex), his inference involved a depth range which was roughly consistent with the range ascribed to *Haskell*'s [1935] inference.

The connection between the Haskell value and the top 1000–1200 km of the mantle clearly also influenced the first applications of spherical viscoelastic Earth models to the GIA problem [e.g., *Cathles*, 1971, 1975; *Peltier*, 1974; *Peltier and Andrews*, 1976]. As an example, *Peltier* [1974] and *Peltier and Andrews* [1976] both considered a suite of models in which the viscosity below 1000 km depth was varied, while the viscosity over the bulk of the top 1000 km was fixed to 10^{21} Pa s.

At this stage an interesting and enduring change in the interpretation of the Haskell value occurred. The motivation for the reinterpretation is somewhat unclear, but there appear to be two contributing issues. First, *Parsons* [1972] considered the resolving power of McConnell's relaxation spectrum. *Peltier et al.* [1978, p. 83] argued on the basis of this work that "McConnell's original data ... are simply incapable of providing any information on the viscosity of the mantle at depths in excess of 600 km." This argument, reiterated in a series of articles [e.g., *Peltier*, 1981, 1982], is not, in fact, supported by the *Parsons* [1972] analysis [see *Parsons*, 1972, Figure 9.9; *Mitrovica and Peltier*, 1993b]. More importantly, the argument seems to have also been applied to the Haskell inference. That is, beginning in the late 1970s, the Haskell value was reinterpreted as being a constraint on the average viscosity in the mantle above the 670-km seismic discontinuity (that is, the upper mantle region in modern terminology). As an example, in their analysis of rotational data, *Yuen et al.* [1982, p. 10,754] wrote that their "calculations ... are based upon an upper mantle (defined to end at 670 km depth) viscosity of (10^{21} Pa s) much cherished in geophysics since *Haskell's* [1935] findings." Over the next decade and a half a large number of influential analyses adopted the same view. For example, *Tushingham and Peltier* [1992, p. 3,285] examined a global database of RSL curves using a suite of Earth models in which the "viscosity of the upper mantle (above 670 km depth) is fixed to the value of 10^{21} Pa s originally deduced by *Haskell* [1935]".

A second contributing cause for the reinterpretation of the Haskell value may have been the emergence of the new, "modern," convention for the radial extent of the upper mantle which places its lower boundary at the 670-km seismic discontinuity. It is possible that over the last two decades the community has simply misinterpreted the upper mantle terminology used in the early papers [e.g., *Anderson*, 1966; *McKenzie*, 1966, 1967; *Anderson and O'Connell*, 1967; *O'Connell*, 1971] as being the same as this modern view.

Whatever the cause of the reinterpretation, there is no doubt that it has greatly influenced research in mantle viscosity. Indeed, a large number of analyses have focused on model spaces in which the upper mantle viscosity was held fixed (with the exception, perhaps, of a low-viscosity zone), at the Haskell value of 10^{21} Pa s, and the lower mantle (below 670 km depth) varied [e.g., *Peltier*, 1982; *Wu and Peltier*, 1983; *Yuen and Sabadini*, 1984; *Mitrovica and Peltier*, 1989; *James*, 1992; *Han and Wahr*, 1995]. There have been suggestions that the average upper mantle viscosity may differ significantly from 10^{21} Pa s [e.g., *Lambeck et al.*, 1990; *Spada et al.*, 1992]. It is interesting to note, however, that these studies are commonly viewed as evidence to dispute the validity of the Haskell value itself [e.g., *Han and Wahr*, 1995] rather than as a suggestion that the "new" upper mantle (above 670 km depth) interpretation of the resolving power of the inference is incorrect.

In the present analysis we determine, for the first time, both the detailed depth dependent averaging ker-

nel associated with the Haskell value and the statistical uncertainty of this average. We begin with a brief outline of the required theoretical formulation and then proceed to a discussion of the results of both forward and inverse analyses of the data set used by *Haskell* [1935]. In this regard we adopt the decay time parameterization of RSL variations advocated by *Mitrovica and Peltier* [1995]. Our goals are to consider whether the modern interpretation of the Haskell value is justified and to provide a rigorous constraint which all inferences of mantle viscosity below Fennoscandia must satisfy. If Haskell's value permits large variations in the upper and lower mantle viscosity (and hence large viscosity contrasts across 670 km depth) but instead constrains the "average" of these viscosities in some sense, then analyses which have assumed that Haskell's value is a constraint on the average value of the upper mantle alone may be significantly biased. Furthermore, if the interpretation of *McKenzie* [1967], *O'Connell* [1971] and others during the same period is closer to the truth, then recent arguments concerning the "contradictory" nature of some inferences of mantle viscosity below Fennoscandia require reappraisal. We will perform this reappraisal using the newly derived averaging kernel. Our reconsideration of the Haskell value of mantle viscosity is thus motivated by both its historical importance and its modern implications.

We will find that the averaging kernel associated with the Haskell value implies a significant level of nonuniqueness in the inference of mantle viscosity. We complete the study by considering whether independent constraints based on decay times associated with RSL variations in the Hudson Bay region (near the center of the massive now vanished Laurentide ice sheet) can reduce this nonuniqueness.

Theory

We adopt a spherically symmetric, self-gravitating, Maxwell viscoelastic Earth model. The gravitationally self-consistent redistribution of ocean meltwater in this case is governed by the so-called sea level equation, which can be written [e.g., *Farrell and Clark*, 1976] as

$$S(\theta, \psi, t) = O(\theta, \psi) [\epsilon(t) + \int_{-\infty}^t \iint_{\Omega} \{\rho_i I(\theta', \psi', t') + \rho_w S(\theta', \psi', t')\} \Phi(\gamma, t - t') d\Omega' dt'] \quad (1)$$

The equation represents a convolution over time, denoted t , and the Earth's surface, denoted Ω , between the total surface load and the potential perturbation Green function Φ [Farrell and Clark, 1976]. In regard to the former, I and S represent the ice and sea level heights, while ρ_i and ρ_w are the densities of ice and water, respectively. The symbol γ represents the angular distance between the load point (θ', ψ') and the observation point (θ, ψ) , where the pairs denote the colatitude and east longitude coordinates, and $\epsilon(t)$ is a spatially invariant time-dependent parameter which is constrained by invoking conservation of mass for the total (ice plus water) surface load. $O(\theta, \psi)$ is the ocean

function, defined to be unity over the oceans and zero elsewhere.

The viscoelastic structure of the Earth model is embedded in the Green function $\Phi(\gamma, t)$. With a single exception described at the end of the paper, we use the normal mode formalism outlined by *Peltier* [1974, 1985] to compute the load Love numbers associated with its construction. The sea level equation is solved using the pseudo-spectral algorithm outlined by *Mitrovica and Peltier* [1991b]. In the present application of the algorithm we adopt a spherical harmonic truncation at degree and order 256, which far exceeds that required for convergence of the Fennoscandian sea level predictions.

RSL variations represent sea level changes relative to the present sea level. Using the solution to (1), we can write

$$RSL(\theta, \psi, t) = S(\theta, \psi, t) - S(\theta, \psi, t_p), \quad (2)$$

where t_p denotes the present time. RSL curves may be generated by solving (1)–(2) for specific geographic sites. The form of these RSL curves is a strong function of the location of the site relative to the ancient ice sheet. As an example, it is well known that in the central portions of previously glaciated regions the RSL curves take the form of rather simple exponential decays [e.g., *Andrews*, 1970; *Cathles*, 1975]. Indeed, *Mitrovica and Peltier* [1993a, 1995] have shown that the equation

$$RSL_i(t) = A_i \{ \exp(t/\Upsilon_i) - 1 \}, \quad (3)$$

where the subscript i denotes the i th geographic site (i.e., $[\theta_i, \psi_i]$), is an accurate model for predicted (and also observed) RSL curves in Hudson Bay if one considers a time range in which the sea level change is dominated by free decay (that is, elastic effects due to any ongoing ice and ocean mass fluctuations are small compared to the nonelastic response to the melting event). In this case, *Mitrovica and Peltier* [1995] demonstrated that the decay times Υ_i associated with RSL variations in this region were relatively insensitive to the details of the late Pleistocene ice history, and they therefore provided a robust data set for the inference of mantle viscosity. The amplitudes A_i are, in contrast, sensitive to the local ice sheet thickness history.

We will use the Monte Carlo approach described by *Mitrovica and Peltier* [1995] to determine the best fitting pair $[A_i, \Upsilon_i]$ for both the predicted and observed RSL variation at each of the two sites considered by *Haskell* [1935]: Angerman River, Sweden, and Oslo, Norway. When applied to the observational record, the Monte Carlo approach permits an estimate of not only the decay time but also its uncertainty.

Our inferences of mantle viscosity will be based on both forward and inverse approaches. In the latter case we apply a standard Bayesian formalism to the inference procedure [e.g., *Backus*, 1971]. The Bayesian approach provides an algorithm for combining observational constraints with a priori information on the model parameters to yield an a posteriori probability distribution for these parameters. *Mitrovica and Peltier* [1995] have demonstrated that the inversion of decay time estimates

from Hudson Bay is rendered weakly nonlinear if the problem is parameterized in terms of the logarithm of viscosity and the logarithm of the decay times. Thus, in describing our inversions, the term “model” will refer to the logarithm of mantle viscosity in a finite number of layers extending from the core-mantle boundary (CMB) to the base of the elastic lithosphere. Furthermore, the term “data” will refer to the logarithm of the decay times determined from the observational constraints. *Tarantola and Valette* [1982] have provided an iterative algorithm for the determination of the maximum likelihood estimate of the a posteriori model distribution, which we denote as X_{PO} , in cases where an extremely large number of forward problems cannot be solved. They have also suggested that the results of linear theory provide an approximate expression for the posterior covariance matrix, which we denote as V_{PO} . The pair $[X_{PO}, V_{PO}]$ represents the solution of our inverse problem, and in particular, we use this solution to determine the value and uncertainty associated with several linear combinations of the a posteriori model parameters (e.g., the average value over the upper mantle).

The determination of the a posteriori model and covariance requires the observational data and its uncertainty, an a priori model and covariance (X_{PR} and V_{PR} , respectively), and Frechet kernels for the data predictions [*Tarantola and Valette*, 1982]. The data will be described in the next section. Inversions will be presented for a suite of prior models; however, in most cases, V_{PR} will be set to $4I$, where I is the identity matrix. That is, no a priori assumptions are made in regard to the smoothness of the model parameters (except on the scale of the layer widths), and the 95% confidence interval for the prior model parameters is given by a $\pm$ eight order of magnitude range about the model.

The Frechet kernels for the data, FK, are defined by

$$\delta \log \Upsilon_i = \int_{CMB/a}^1 FK[\nu(r), r] \delta \log \nu(r) r^2 dr, \quad (4)$$

where r is the radius (nondimensionalized using the mean radius of the Earth, a) and $\nu(r)$ is the profile of mantle viscosity. The nonlinearity of the problem, although weak, is made explicit by the dependence of the Frechet kernels on $\nu(r)$. These kernels relate arbitrary perturbations in the radial viscosity profile to the consequent perturbation in the decay times, and they provide a direct measure of the detailed depth-dependent sensitivity of the predictions to variations in mantle rheology. We compute the kernels using the numerical perturbation procedure outlined by *Mitrovica and Peltier* [1993c]; in this regard, we discretize the sublithospheric mantle into 32 layers, 22 of which reside in the lower mantle (i.e., below 670 km depth).

Results

Observational Data

Following *Haskell* [1935], we will consider RSL variations at two sites within the Fennoscandian region:

Angerman River, Sweden, and Oslo, Norway. The RSL curve for Angerman River has played a particularly prominent role in previous inferences of mantle viscosity [e.g., *Walcott*, 1973; *Peltier*, 1982; *Wolf*, 1986, 1987; *Lambeck et al.*, 1990], and a number of workers have commented on its apparently excellent reliability [e.g., *Walcott*, 1973; *Peltier*, 1982; *Lambeck et al.*, 1990; *Ekman*, 1993]. The observational record for this site is based on the raised beach observations of *Liden* [1938], although a number of corrections are necessary to relate the varve chronology associated with this data to the modern C^{14} time scale (see *Lambeck et al.* [1990] for details). We adopt the corrected data set presented by *Lambeck et al.* [1990], which is shown in Figure 1a after correction to the sidereal timescale.

Following *Lambeck et al.* [1990], the age standard deviation for each of the Angerman River RSL data points is taken to be 250 years. The height datum associated with the Angerman River data is uncertain, and for this reason we incorporate the possibility of a constant datum shift into our analysis. In particular, following *Lambeck et al.* [1990], we amend equation (3), when it is applied to the observational record, by adding a site-dependent constant (which we will denote as C_i) that is simultaneously estimated during the Monte Carlo procedure.

The observational constraints for Oslo, Norway, are shown in Figure 1b. These data are taken from *Hafsten* [1979] and the reader is referred to this article for

details. Once again, we will permit the possibility of a constant datum shift in fitting the form (3) to the observational constraints for this site.

The RSL data for Angerman River and Oslo exhibit the exponential-like form associated with previously glaciated regions. In the next section, once we have established an appropriate time window for the analysis, we will consider the accuracy of the form (3) applied to these data. We will also estimate the decay times and their uncertainty that are associated with the Monte Carlo derived best fit.

Our analysis will be based on the actual data obtained by survey (Figures 1a and 1b) rather than the “trends” adopted in many previous analyses. As an example, in both the *Lambeck et al.* [1990] analysis of RSL variations at Angerman River and the *Tushingham and Peltier* [1992] analysis of RSL variations in Oslo, a subjectively chosen envelope was drawn around the actual data set. A second “data set,” specified at a number of integer times, was then derived to reflect the trend of the envelope. The latter constraints were subsequently used in the inference of mantle viscosity. As discussed by *Mitrovica and Peltier* [1995], this approach may be problematic, particularly if large temporal gaps exist in the raw RSL data (as in the case of Oslo, Figure 1b).

Finally, in the *Lambeck et al.* [1990] analysis, errors in the age determinations were mapped into pseudo-errors in beach heights using the slope of the envelope trend. Our analysis of the observed RSL record, based on decay times derived using a Monte Carlo procedure, does not require such a mapping.

Forward Analyses

The results to be described in this section are based on Earth models with an elastic structure given by the seismic model PREM [*Dziewonski and Anderson*, 1981], an elastic lithosphere, and uniform upper and lower mantle viscosities (which we denote by ν_{UM} and ν_{LM} , respectively). We will consider two models for the space time history of the late Pleistocene ice sheets. The first is a highly simplified disk model of the Fennoscandian ice complex (henceforth the “DISK” ice load). The model is characterized by a single disk of circular horizontal cross section, parabolic vertical cross section, and a mass which varies linearly with time during a deglaciation phase which extends from 18 to 9 kyr before present (B.P.). The disk is assumed to maintain a plastic equilibrium profile [*Patterson*, 1981] during this period and to have a radius of 8° and a height of 2500 m at the onset of the deglaciation event. The second ice model is an adaptation of the ICE-1 model of *Peltier and Andrews* [1976]. In particular, the model is discretized to a finer spatial resolution than the 5° disks which characterized the original model, and it is augmented to include an Antarctic melting event. In all cases the Earth model is assumed to be in a state of isostatic equilibrium at the onset of the deglaciation event. *Mitrovica and Peltier* [1995] have shown that departures from this state do not influence estimates of postglacial decay times for RSL curves in Hudson Bay, and we have confirmed this insensitivity using our Fennoscandian database.

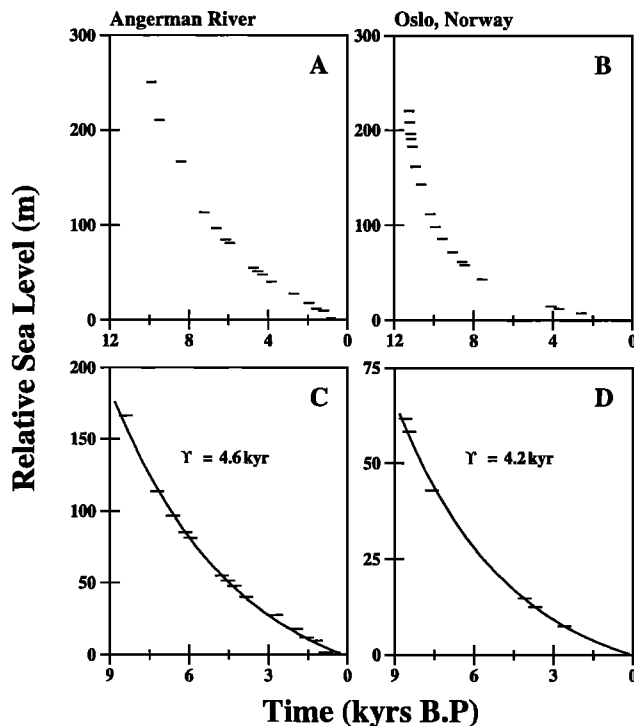


Figure 1. Observational constraints on RSL variations at (a) Angerman River, Sweden, and (b) Oslo, Norway, over the last 12 kyr. The best fitting form (3) (augmented by a site-dependent datum shift) for RSL variations over the last 9 kyr at (c) Angerman River and (d) Oslo. The value of τ plotted in Figures 1c and 1d represents the decay time of this best fitting form.

In estimating characteristic decay times from RSL variations at Angerman River and Oslo we require a time window in which the single-exponential form (3) is valid. To obtain a conservative estimate of this time range we have adopted the following approach, illustrated by the results in Figures 2a and 2b. First, predictions of RSL variations at Angerman River and Oslo were computed over the last 16 kyr. Next, a Monte Carlo search was performed to determine the best fitting form (3) through the RSL predictions over the last 5 kyr only. This search yielded decay time estimates of 5.2 kyr and 4.7 kyr, for Angerman River and Oslo, respectively, in the case of the RSL predictions shown in Figures 2a and 2b. The dashed lines in Figures 2a and 2b represent these best fitting forms plotted over the last 16 kyr. In spite of the fact that a 5-kyr time window was used in the derivation of the decay time estimates (the 5-kyr period was chosen because the dominance of free-decay adjustment during this period, over Fennoscandia, is assured), the single-exponential

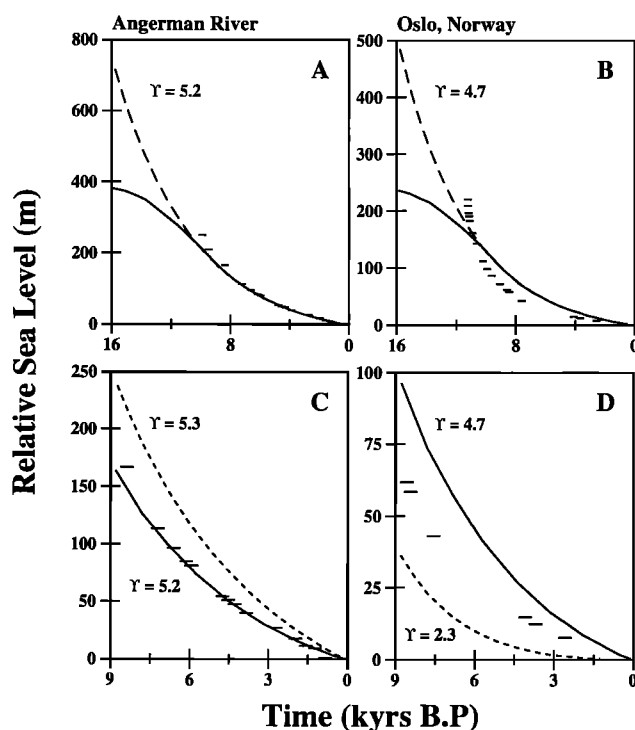


Figure 2. Numerical predictions (solid lines) of RSL variations at (a) Angerman River and (b) Oslo based on the deglaciation history adapted from the ICE-1 model and an Earth model characterized by a 120-km elastic lithosphere and an isoviscous, 10^{21} Pa s, mantle. The dashed lines represent the best fitting form (3) determined using only the most recent 5 kyr of the predictions, and the value of τ labeled on each plot is the decay time associated with this form. Predicted RSL variations at (c) Angerman River and (d) Oslo based on the same Earth model used in Figures 2a and 2b and either the ICE-1 deglaciation history (solid lines) or the DISK ice load history (dotted lines). The values of τ labeled on each frame are the decay times of the best fitting form (3) through the adjacent (9 kyr) numerical predictions.

parameterization of the RSL predictions is essentially indistinguishable from the the RSL curves generated using the full multinormal mode forward theory (1)-(2) over the last 10 to 11 kyr. We have repeated these calculations for a suite of Earth models and ice histories and have found that the accuracy of the single exponential form (3) over the last 10 kyr is a robust feature of the RSL predictions at our two sites. However, to be even more conservative, we adopt, in the remainder of this study, a 9-kyr time window.

The solid lines in Figures 2c and 2d represent the portion of the (multinormal mode) predictions plotted in Figures 2a and 2b which fall within our chosen 9-kyr time window. The decay times determined from these 9-kyr time series agree to within 1% with those obtained using the 5-kyr time range described above. This is yet another indication of the constancy of the best fitting decay time over this longer time period.

With the time window established, we can return to the observational record described in the last section. In Figures 1c and 1d we plot the best fitting form (3) (augmented by the site-dependent datum adjustment C_i described above) through the uplifted beach records over this time range. The decay times generated from the Monte Carlo search, 4.6 ± 0.8 kyr for Angerman River and 4.2 ± 0.7 kyr for Oslo, are consistent with each other. Furthermore, even a cursory examination of Figures 1c and 1d indicates that the single-exponential form (3) represents an exceedingly accurate model for the observed RSL record over the last 9 kyr at these two sites. Indeed, the χ^2 residual per degree of freedom between the best fitting model (3) and the observational record is just 0.29 for Angerman River and 0.10 for Oslo.

The Monte Carlo search applied to the observational record yielded constant shifts in the height datum of -1.2 m for Angerman River and 0.1 m for Oslo. Neither of these values is different from zero at 95% statistical confidence, and the former is consistent with the range of values cited by *Lambeck et al.* [1990] for the Angerman River site.

In Figures 2c and 2d we examine the sensitivity of the decay time predictions to variations in the adopted ice model. The solid lines were based on the ICE-1 deglaciation history, while the dotted lines were computed using the DISK model. The discrepancies in the RSL predictions associated with the two ice models far exceed the observational uncertainties, and hence inferences of viscosity based upon the raw RSL database at these two sites would be significantly biased by errors in the ice-load history. The same is not necessarily true of inferences based on the decay time estimates. In particular, the decay times of the RSL predictions at Angerman River generated using the two ice histories differ by less than 2% (Figure 2c). This discrepancy is significantly less than the uncertainty in the observed decay time at this site (Figure 1c). Unfortunately, this remarkable insensitivity does not extend to the Oslo site. In this case the decay times based on the two ice histories are discrepant by a factor of two (Figure 2d). The robustness of the decay time estimate is clearly a

function of the location of the site relative to the ancient Fennoscandian ice complex. Oslo's proximity to the edge of the ice sheet increases its sensitivity to the details of the deglaciation event.

In Figure 3 we explore the sensitivity of the decay time estimates to variations in the lithospheric thickness of the Earth model. Predictions are shown for a range of lower mantle viscosities (as labeled on the abscissa) and two different lithospheric thicknesses: 70 km and 120 km. The decay time estimates associated with the Angerman River site, which is situated near the central peak of the Fennoscandian ice complex, are insensitive to the adopted lithospheric thickness [see also *Cathles, 1975*]. Indeed, the decay time estimates for this site based on the "thin" lithosphere model are, on average, only 0.2 kyr greater than those computed using the Earth model with a 120-km lithosphere. In contrast, Oslo's location relative to the Fennoscandian ice complex introduces a much more pronounced sensitivity to lithospheric thickness.

We conclude that the characteristic decay time of the RSL record at Angerman River provides a robust constraint on mantle viscosity which is insensitive to errors in both the assumed melting history of the Fennoscandian ice complex and the adopted lithospheric thickness of the Earth model. In the remainder of this section and the next we focus on this site. As an example, in Figure 4a we show predictions of the decay time at

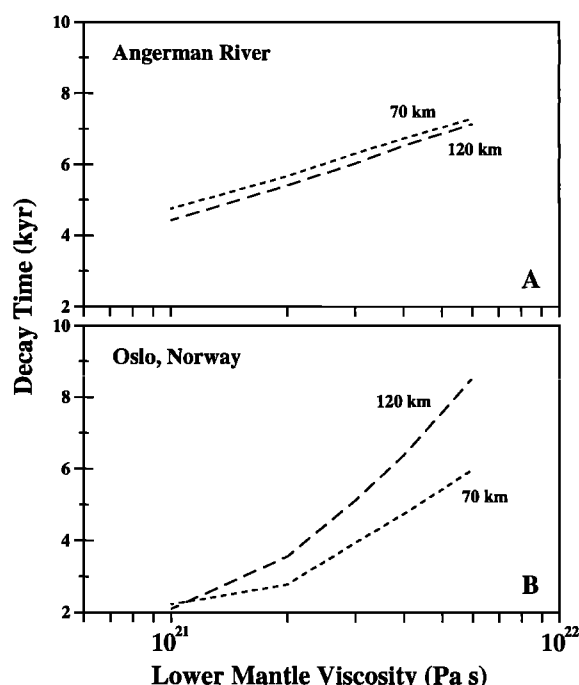


Figure 3. Decay times associated with the best fitting form (3) through numerical predictions of RSL variations over the last 9 kyr at (a) Angerman River and (b) Oslo computed using the DISK ice load history and a suite of Earth models. These models are distinguished on the basis of ν_{LM} (as labeled on the abscissa). Furthermore, on each frame, results are presented for lithospheric thicknesses of 70 km and 120 km. In all cases $\nu_{UM} = 7 \times 10^{20}$ Pa s.

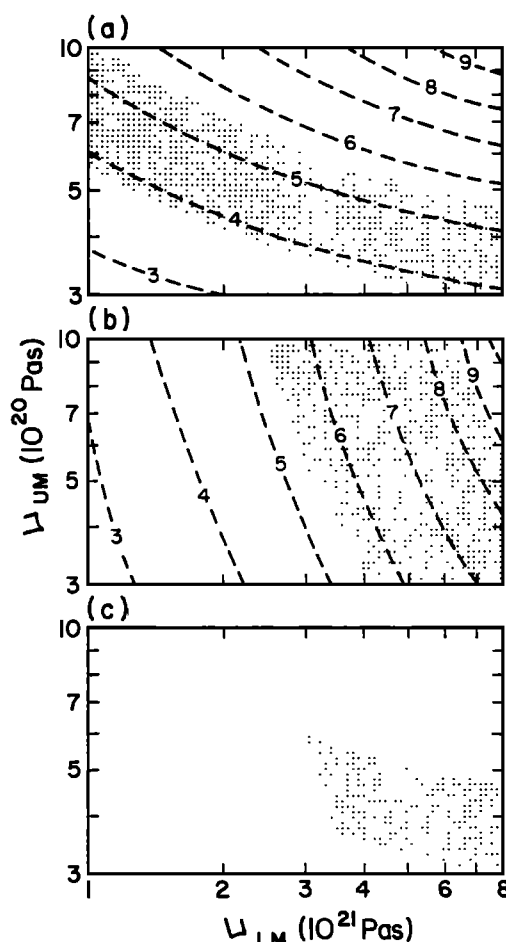


Figure 4. Decay times (units of kyr) associated with the best fitting form (3) through numerical predictions of RSL variations (a) over the last 9 kyr at Angerman River and (b) over the last 6.5 kyr at Richmond Gulf. The predictions are based on the DISK ice load histories and a suite of Earth models defined by varying ν_{UM} and ν_{LM} . The lithospheric thickness is fixed at 120 km. The shaded regions reflect the observational constraint at each site. (c) The intersection of the preferred model spaces of Figures 4a and 4b.

Angerman River based on a suite of Earth models in which ν_{UM} and ν_{LM} are varied over ranges which are commonly cited in the literature.

The results in Figure 4a are significant in several respects. First, they indicate that the Angerman River RSL data are sensitive not only to the upper mantle viscosity but also to variations in viscosity below 670 km depth (this sensitivity is also evident in Figure 3a). Haskell's [1935] inference of viscosity, if it is correct, must therefore be associated with a radial region which extends into the lower mantle. As we discussed in the introduction, a large number of analyses over the last two decades have adopted the Haskell value (10^{21} Pa s) as a constraint on the average viscosity of the upper mantle alone. We conclude that these analyses have misinterpreted the resolving power of the constraint and have therefore introduced potentially significant bias into their conclusions. As an example, if one fixes ν_{UM}

to 10^{21} Pa s, then the Angerman River decay time is satisfied by a model in which $\nu_{LM} \sim 0.6 - 1.1 \times 10^{21}$ Pa s. However, this decay time is equally well satisfied by a model characterized by $\nu_{UM} = 3 \times 10^{20}$ Pa s and $\nu_{LM} = 8 \times 10^{21}$ Pa s; that is, a model with greater than a factor of 25 jump in viscosity across 670 km depth.

The correlation between preferred values of ν_{UM} and ν_{LM} stays relatively constant over the range of Earth models considered in Figure 4a, although the gradual reduction in slope indicates, not surprisingly, that the sensitivity to variations in ν_{LM} decreases as that region is stiffened. This near-constancy reflects a quasi-linear dependence of the predictions to variations in the logarithm of mantle viscosity, and this property will be key to ensuring rapid convergence of the inversions described below.

The results in Figure 4a represent a rather coarse discretization of the detailed depth-dependent sensitivity of the Angerman River decay time estimate to variations in mantle viscosity. Although a sensitivity to lower mantle viscosity is indicated in Figure 4a, it is not clear how deep this sensitivity extends into the lower mantle. That is, the resolving power of the data, or the “averaging” depth implied by Haskell’s analysis, has not been established. In the next section we derive a Frechet kernel for the Angerman River prediction which provides this information. We furthermore establish a bound on the uncertainty of this average which unifies the ambiguous constraint on mantle viscosity implied by the results in Figure 4a.

Inverse Analyses

The Frechet kernel (equation (4)) for a prediction of the decay time at Angerman River, based on an Earth model characterized by an isoviscous (10^{21} Pa s) mantle, is shown in Figure 5a.

Peltier [1976], following the original analysis of Parsons [1972], has derived analytic expressions for Frechet kernels relating perturbations in the logarithm of the mantle viscosity to the consequent perturbation in the logarithm of the decay times of the individual normal modes of viscous gravitational relaxation. These kernels have two important characteristics: They are everywhere greater than zero; and their integral, over Earth radius, is unity. That is,

$$\int_{CMB/a}^1 FK[\nu(r), r] r^2 dr = 1. \quad (5)$$

These properties are shared, at least in an approximate sense, by our numerically derived kernel for the decay time associated with the full waveform (i.e., lumped multinormal mode) RSL variation at Angerman River. The sign of this kernel in Figure 5a indicates, as one might expect, that a positive perturbation in the viscosity at any location within the mantle always leads to an increase in the characteristic decay time of the RSL curve. Furthermore, the integral in (5) applied to the Angerman River kernel in Figure 5a yields the value 0.8. This is close to the analytic result (5) for the individual normal modes, and it suggests that an order of

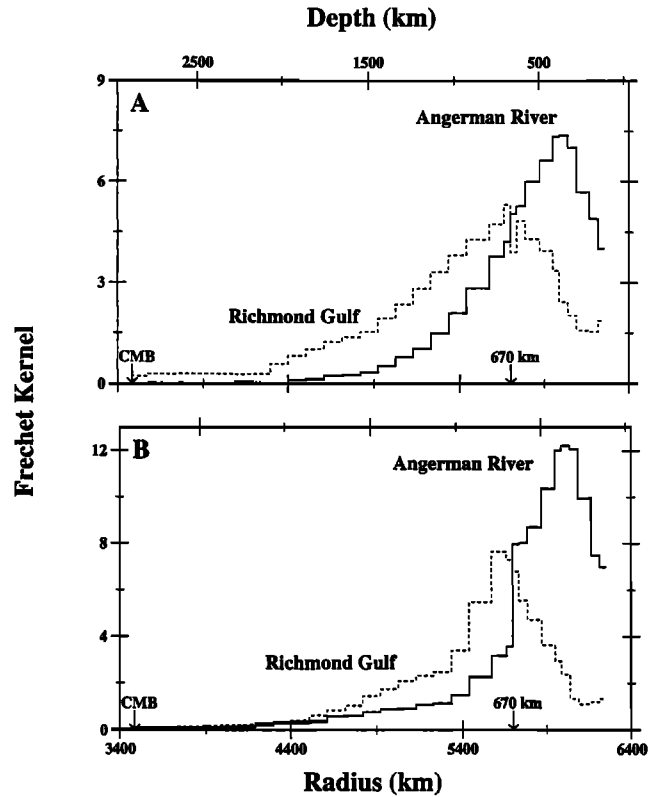


Figure 5. Frechet kernels (equation (4)), scaled by the square of the dimensionless radius (i.e., $FK(r)r^2$), for numerical predictions of the decay time associated with RSL variations over the last 9 kyr at Angerman River, Sweden (solid lines), and over the last 6.5 kyr at Richmond Gulf, Quebec (dotted lines). The predictions are based on the DISK ice load histories, an Earth model with a lithospheric thickness of 120 km and (a) $\nu_{UM} = \nu_{LM} = 10^{21}$ Pa s, or (b) $\nu_{UM} = 5 \times 10^{20}$ Pa s and $\nu_{LM} = 5 \times 10^{21}$ Pa s.

magnitude increase in the viscosity of the mantle will lead to approximately an order of magnitude increase in the predicted decay time. These agreements suggest a consistency between the physics of the glacial isostatic adjustment process and the decay time parameterization.

The area under the kernel for the Angerman River decay time in Figure 5a is partitioned in the ratio 2:1 within the upper and lower mantle regions. That is, a given perturbation in the logarithm of ν_{UM} will produce double the perturbation in the logarithm of the decay time than an equal logarithmic perturbation in ν_{LM} . This is consistent with the slope of the contours in Figure 4a through the model used to compute the kernel ($\nu_{UM} = \nu_{LM} = 10^{21}$ Pa s), and it indicates, once again, that the data from central Fennoscandia have a nonnegligible sensitivity to lower mantle viscosity.

The decay time associated with the RSL variation at Angerman River has a sensitivity (Figure 5a) which peaks at mid-upper mantle depths. The sensitivity diminishes rapidly at shallower depths, which is consistent with our earlier observation that the decay time prediction is relatively insensitive to variations in the

lithospheric thickness. Furthermore, the sensitivity remains nonnegligible to a depth of 1400 km. We conclude that the lower mantle sensitivity implied by the results in Figures 3 and 4a is actually limited to the top ~600–700 km of this region. The decay time prediction provides little constraint on the viscosity in the bottom half of the mantle.

These results indicate that the early interpretation of Haskell's value as a constraint on the viscosity in the top 1000–1200 km of the mantle [e.g., McKenzie, 1966, 1967; O'Connell, 1971] is essentially correct. We conclude once again that the more recent (re)interpretation of the Haskell value as a constraint on the average viscosity of the upper mantle alone (see the introduction) is in general incorrect and has likely led to significant bias in inferences of mantle viscosity and other studies related to glacial isostatic adjustment.

The quasi-linear nature of the results in Figure 4a suggests that our interpretation of the Haskell constraint remains valid over a wide range of Earth models (and viscosity contrasts across 670 km depth). To consider this issue we plot, in Figure 5b, the Frechet kernel for the Angerman River decay time computed using an Earth model characterized by $\nu_{UM} = 5 \times 10^{20}$ Pa s and $\nu_{LM} = 5 \times 10^{21}$ Pa s. As before, the kernel is everywhere positive and has an integrated area near unity. The ratio of the area under the kernel within the upper and lower mantle regions is now approximately 3:1. This is also consistent with the results in Figure 4a, and it indicates, not surprisingly, that the sensitivity of the datum to variations in viscosity becomes more focused within the upper mantle as the viscosity contrast across 670 km depth increases. The trend is, however, sufficiently weak (that is, the relationship between logarithmic perturbations in the decay times and viscosity is sufficiently linear) that the sensitivity of the datum to variations in viscosity down to depths of about 1400 km depth remains important even in the case of an order of magnitude jump in viscosity across the 670-km interface.

In the remainder of this section we consider a sequence of Bayesian inversions based on the Angerman River decay time constraint. As an example, Figure 6a shows the posterior model for an inversion generated using a prior model characterized by a uniform value of 21.0 (recall, our inversions are parameterized in terms of the logarithm of viscosity). Since the Angerman River decay time provides little information in regard to rheology below the top portion of the lower mantle, the posterior model tends toward the prior model in this depth range (i.e., below about 1000 km depth). Above this depth the posterior model values are slightly lower than the prior values, and we will quantify this observation using several "averages." In particular, let us define a general linear combination of the posterior model parameters as:

$$A = \frac{\int_{CMB/a}^1 W(r) X_{PO} r^2 dr}{\int_{CMB/a}^1 W(r) r^2 dr}. \quad (6)$$

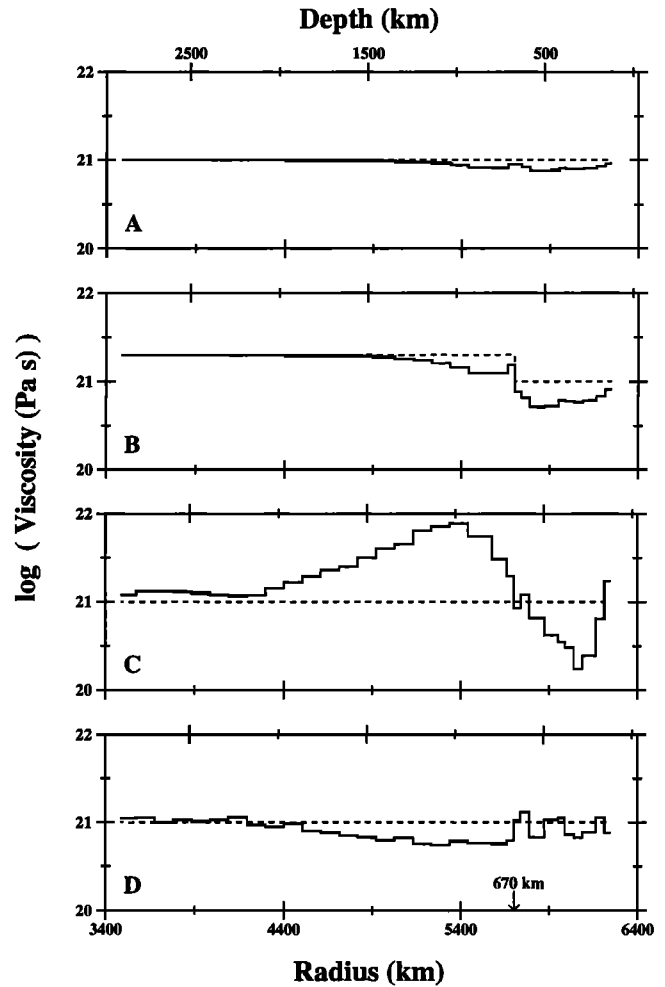


Figure 6. (a,b) Prior (dotted lines) and posterior models (solid lines) for the inversion of the decay time associated with RSL variations at Angerman River, Sweden. (c) As in Figures 6a and 6b, except the inversion is based on the decay times associated with RSL curves at both Angerman River and Richmond Gulf, Quebec. (d) As in Figures 6a and 6b, except the inversion is based on the decay times associated with RSL curves at Angerman River, Southampton Island, Ottawa Island, and Ungava Peninsula. All numerical predictions are based on the DISK ice load histories and an Earth model with a lithospheric thickness of 120 km.

The volumetric average of the model parameters in the upper mantle (which we denote by A_{UM}) is defined by setting $W(r) = 1$ in the upper mantle and zero elsewhere. Furthermore, a linear combination of the model values weighted by the sensitivity implied by the Angerman River Frechet kernels in Figures 5a and 5b (denoted as A_{AR}^A and A_{AR}^B , respectively) may be defined by setting the averaging kernel $W(r)$ to the appropriate function $FK(r)$. The estimates A_{UM} and A_{AR}^A are 20.91 ± 0.35 and 20.92 ± 0.10 , respectively, for the case of the posterior model in Figure 6a. (Since this posterior model differs only moderately from the isoviscous, 10^{21} Pa s, prior model, the averaging kernel A_{AR}^A is more appropriate than A_{AR}^B .)

The smaller uncertainty in the average A_{AR}^A , relative to the uncertainty in A_{UM} , reflects the fact that the resolving width of the Angerman River decay time (Figure 5a) extends over a significantly broader range than the upper mantle. This (lack of) resolution is also evident in the posterior correlations associated with the estimates. If we denote a bulk lower mantle average as A_{LM} ($W(r) = 1$ in the lower mantle and zero elsewhere), then we have found that the correlation coefficient between the posterior estimates for A_{UM} and A_{LM} is -0.60 (as one would roughly expect from Figure 4a), while the correlation coefficient between the estimates for A_{AR}^A and A_{LM} is just 0.07 .

The robustness of the posterior estimate A_{AR}^A is also evident in Figures 6b and 7, where we consider the effect of varying the lower mantle value of the prior model. In Figure 6b these values are set to 21.3 (or 2×10^{21} Pa s). In this case, while the posterior estimate for A_{UM} is reduced considerably, the posterior estimate for A_{AR}^A remains essentially unaltered at 20.92 ± 0.10 . We verify, in Figure 7, that the latter remains approximately valid over a wide range of prior models.

The posterior estimates shown in Figure 7 should, strictly speaking, be determined using Frechet kernels (i.e., averaging functions) computed on the basis of the posterior model obtained from each inversion. We have found, as a consequence of the quasi-linear nature of the inversions, that the model dependence of the averaging kernel does not significantly alter the posterior constraints. As an example, consider results analogous to those of Figure 7 for the average A_{AR}^B defined on the basis of the Angerman River decay time kernel in Figure 5b. This average ranges from 20.92 to 20.86 (with a standard deviation near 0.13) as the lower mantle value of the prior model increases from 21.0 to 21.7 . In

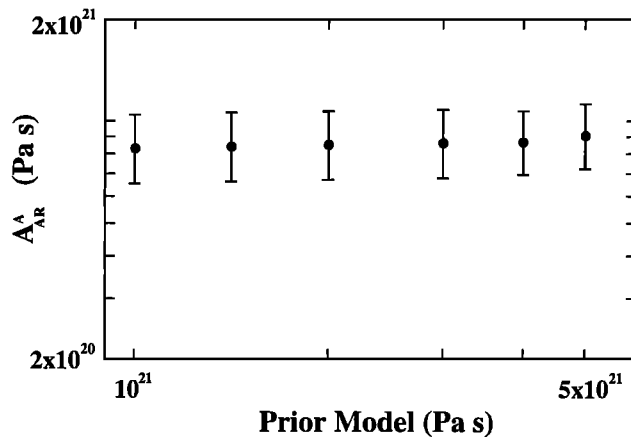


Figure 7. The posterior average A_{AR}^A (see text) for a series of inversions of the Angerman River decay time. These inversions are distinguishable on the basis of the prior model adopted for the lower mantle (as shown on the abscissa scale). The vertical bars represent the $\pm 1\sigma$ uncertainty in the posterior estimate. All numerical calculations are based on the DISK ice load history and an Earth model with a lithospheric thickness of 120 km. The prior value of the model parameters in the upper mantle is fixed at 21.0 (or 10^{21} Pa s on a linear scale).

the remainder of this section we focus, exclusively, on results based on the average A_{AR}^A with the caveat that there will be a slight sensitivity to the choice of averaging kernel.

The posterior constraint on A_{AR}^A , which refers to a viscosity range of roughly $0.65 - 1.1 \times 10^{21}$ Pa s, is in agreement with the value derived by Haskell [1935]. We conclude that Haskell's inference is accurate but that it must be associated with the averaging kernel defined by the Frechet kernel for the Angerman River decay time in Figure 5a. This average is remarkably robust and unbiased. As we have demonstrated, it is insensitive to variations in the late Pleistocene ice history and the choice of lithospheric thickness. It also permits large variations in viscosity across 670 km depth.

The latter is well illustrated by applying the averaging function to a number of apparently contradictory inferences of mantle viscosity based on the GIA data set from Fennoscandia [McConnell, 1968; Wolf, 1987; Lambeck et al., 1990; Fjeldskaar and Cathles, 1991]. Wolf [1987] analyzed RSL data from Angerman River and Helsinki using a mantle model with three layers; an elastic lithosphere of variable thickness, a 100 -km sublithospheric zone of variable (but relatively low) viscosity, and (motivated by the Haskell result) a 10^{21} Pa s substratum. He derived an upper bound on the thickness of the Fennoscandian lithosphere of 80 km. Fjeldskaar and Cathles [1991] used a similar three-layer model but allowed the viscosity of the mantle substratum to vary in analyzing Fennoscandian data. They inferred a mantle viscosity of 1.2×10^{21} Pa s. In contrast to these studies, Lambeck et al. [1990] used a three-layer mantle model, defined by an elastic lithosphere and isoviscous upper and lower mantle regions (the boundary between the two was taken to be 670 km depth) to analyze RSL curves at four sites in NW Europe. The RSL curve for Angerman River played an important role in their analysis, and they derived the following parameter ranges: lithospheric thickness 100 - 150 km; upper mantle viscosity $(3 - 5) \times 10^{20}$ Pa s; and lower mantle viscosity $(2 - 7) \times 10^{21}$ Pa s. They furthermore ruled out an isoviscous 10^{21} Pa s mantle.

If we apply the newly derived Haskell averaging kernel A_{AR}^A to the preferred viscosity models of McConnell [1968], Wolf [1987], Lambeck et al. [1990] (in particular, the model $\nu_{UM} = 3 \times 10^{20}$ Pa s and $\nu_{LM} = 7 \times 10^{21}$ Pa s), and Fjeldskaar and Cathles [1991], then we obtain the remarkably consistent values 20.88 , 20.93 , 20.93 , and 20.98 , respectively. All of these fall well within the bound we have derived for this average, and accordingly, each of the models yields a characteristic decay time for the Angerman River RSL variation which satisfies the observational constraint. Indeed, while these models range from having no viscosity contrast across 670 km depth to having a factor of 23 jump across this interface, they yield decay times in the relatively small range 4 to 5 kyr.

This agreement suggests, but does not prove, that the apparently contradictory nature of these viscosity inferences may be the consequence of the different man-

the discretizations adopted by these authors. Since they adopt an isoviscous bulk mantle, *Fjeldskaar and Cathles* [1991] would require a viscosity near 10^{21} Pa s to satisfy the decay time constraint from central Fennoscandia. This is not true of the upper mantle/lower mantle discretization used by *Lambeck et al.* [1990]; in this case, their inferred viscosity jump at 670 km must be forced by other data used in their analysis. A final word on this issue requires an inversion of the complete data sets adopted by these authors, but there are two points which tend to support our suggestion. First, the correlation between the upper and lower mantle viscosities preferred by *Lambeck et al.* [1990] (they found that the higher the lower mantle, the weaker the preferred upper mantle) is entirely consistent with the preferred model space given by the shaded region in Figure 4a. Second, *Fjeldskaar* [1994] has argued that his preferred lower mantle viscosity can increase to 2×10^{21} Pa s if the bulk upper mantle viscosity is reduced to 7×10^{20} Pa s. Significantly, this change leaves the average cited above for the *Fjeldskaar and Cathles* [1991] model essentially unaltered.

As a final point, the model suggested by *Tushingham and Peltier* [1992] in their analysis of a global RSL database (including Angerman River and Oslo; $\nu_{UM} = 10^{21}$ Pa s, $\nu_{LM} = 2 \times 10^{21}$ Pa s) yields a decay time (~ 6.5 kyr) which is somewhat too large to reconcile the observational constraint at Angerman River. This is an illustration of the fact that the “true” Haskell constraint can rule out a significant class of Earth models (see Figure 4a) and certainly does not imply that $\nu_{UM} = 10^{21}$ Pa s.

As discussed in the theory section, our calculation of the surface load Love numbers are based on the impulse response multinormal mode formalism outlined by *Peltier* [1974, 1985]. There has been some recent suggestion [*Fang and Hager*, 1994] that this theory may not yield a complete representation of the impulse response of a Maxwell viscoelastic model when continuous viscosity variations with depth are considered. We have implemented a version of the CRFT approach advocated by *Fang and Hager* [1994] and have found that our constraint on the “average” mantle viscosity associated with the Angerman River decay time (A_{AR}^A) is essentially unchanged (20.93 ± 0.11).

Decay Time Constraints at Richmond Gulf and Northern Hudson Bay

The nonuniqueness in estimating, separately, the upper and lower mantle viscosity from the Angerman River decay time can only be reduced by applying independent constraints to the inference. To complete this study, we provide preliminary examples of this approach based on invoking decay time estimates from the second major region of late Pleistocene deglaciation: Hudson Bay. *Mitrovica and Peltier* [1995] have shown that decay times determined from RSL variations over the last 6.5 kyr in the Hudson Bay region can provide a robust (in the sense we have used it above) constraint on mantle viscosity in a region extending from the tran-

sition zone down to about 2000 km depth. The RSL data set in this region is sparse and of uneven quality. In particular, while *Mitrovica and Peltier* [1995] considered six sites, three in northern Hudson Bay and three in southern Hudson Bay, there is likely just one RSL curve, at Richmond Gulf, Quebec, with an accuracy comparable to that of the Angerman River data. The main reason for this is that the Richmond Gulf RSL variation [*Hillaire-Marcel*, 1980] is constrained by many more data points than others from the region within the adopted 6.5-kyr time window (eight compared to only three or four age-height pairs for the other sites [*Tushingham and Peltier*, 1991]). Furthermore, the Richmond Gulf RSL curve is, unlike some other sites, based on a relatively consistent set of age-height samples.

The location of Richmond Gulf, in southern Hudson Bay, is also significant. Sites in northern Hudson Bay are in proximity to the edge of the craton, and may thus be significantly influenced by lateral heterogeneities associated with the margin [e.g., *Sabadini et al.*, 1986; *Gasparini and Sabadini*, 1989; *Manga and O’Connell*, 1995]. Furthermore, southern Hudson Bay is close to the center of the ancient Laurentide ice sheet at last glacial maximum. Therefore predicted decay times for the Richmond Gulf site will be less sensitive to the details of the ice load history than the decay times associated with sites in northern Hudson Bay.

Han and Wahr [1995] have recently suggested that the curvature of some RSL curves in Hudson Bay can be altered by considering a multidome rather than single dome model of the Laurentide ice complex. However, the time window used in their estimate of curvature was not limited to the postglacial (i.e., predominantly free decay) period. Inspection of their results (see their Figure 15) indicates that characteristic decay times over the last 6.5 kyr, particularly for sites in southern Hudson Bay (e.g., Ottawa Island, Cape Henrietta Maria), are remarkably insensitive to the large variations in ice sheet models which they consider. Indeed, we have determined the best fitting form (3) through the *Han and Wahr* [1995, Figure 15, dashed and dotted lines] RSL predictions at Ottawa Island and Cape Henrietta Maria over the last 6.5 kyr, and have found that the decay times at each site vary by less than 2% for the two ice histories. We have verified this insensitivity for our own analogous predictions at Richmond Gulf.

In Figure 4b we consider predictions of the decay time at Richmond Gulf based on a suite of Earth models distinguished by the adopted values of ν_{UM} and ν_{LM} . The calculations adopt a DISK ice model characterized by a maximum radius of 16° , a peak central height of 3200 m, and a deglaciation event extending from 16 to 7 kyr B.P. A comparison with Figure 4a indicates that the decay time estimate at Richmond Gulf is significantly more sensitive to bulk variations in the lower mantle viscosity than the estimate for the Angerman River site is. This is, of course, as one would expect given the relative sizes of the Fennoscandian and Laurentian ice complexes. The results also indicate that the Richmond Gulf decay time cannot be reconciled with a value of

ν_{LM} less than approximately 2.4×10^{21} Pa s [see also *Mitrovica and Peltier, 1995*].

In Figure 4c we consider the space of our two-layer (ν_{UM}, ν_{LM}) models which satisfy the observational constraints on the decay times at both Angerman River and Richmond Gulf. This space is roughly defined by a lower bound on ν_{LM} of 3×10^{21} Pa s and the range $3 \times 10^{20} \leq \nu_{UM} \leq 6 \times 10^{20}$ Pa s. Significantly, this inference is in good agreement with the model space inferred by *Lambeck et al. [1990]* on the basis of GIA data from NW Europe. The smallest jump in viscosity across 670 km depth associated with the model space in Figure 4c is a factor of 5; however, contrasts of an order of magnitude or more are not ruled out.

Frechet kernels for the prediction of the Richmond Gulf decay time are shown in Figure 5. The kernel computed using an Earth model having an order of magnitude increase in viscosity across the 670 km interface (dotted line, Figure 5b) has a lower mantle sensitivity which is more focused within the shallowest portion of that region than the kernel for the same site based on the isoviscous (10^{21} Pa s) mantle (Figure 5a). Nevertheless, the ratio of the areas under the Richmond Gulf decay time kernels within the upper and lower mantle is roughly 1:2 in both calculations, which is consistent with the relatively gentle change in the slope of the contours in Figure 4b.

The amplitudes of the Richmond Gulf kernels remain nonnegligible to a depth of about 2000 km, which is significantly deeper than the depth penetration of the Angerman River kernels. The Richmond Gulf decay time is also less sensitive to variations in upper mantle viscosity; recall that about 70% of the total area under the Angerman River kernels resides in this region (compared to 30% in the Richmond Gulf case). The increase in deep mantle sensitivity, at the expense of upper mantle sensitivity, indicates (as implied by Figure 4c) that an inversion of both the Richmond Gulf and Angerman River decay times will be characterized by less nonuniqueness in regard to the viscosity contrast across the upper and lower mantle regions than were our previous inversions (Figures 6a, 6b, and 7).

In Figure 6c we show the results of a simultaneous inversion of the decay time constraints from these two sites. The predicted decay times for both the prior and posterior models are shown in Figure 8. Inspection of Figure 8 indicates that the prior model (characterized by an isoviscous 10^{21} Pa s mantle) fits the observational constraint at Angerman River at the level of one standard deviation but that it significantly misfits the decay time estimated for the Richmond Gulf RSL variation (see Figure 4b). Indeed, the χ^2 residual between the observational constraints and predictions based on the prior model is 9.3. In contrast, the χ^2 residual for the fit of the posterior model is just 0.4. This improvement in the fit is due primarily to the longer decay time at Richmond Gulf predicted on the basis of the posterior model (6.0 kyr compared to 3.3 kyr for the prior model).

The resolving power of the inversion shown in Figure 6c originates from the sensitivity associated with

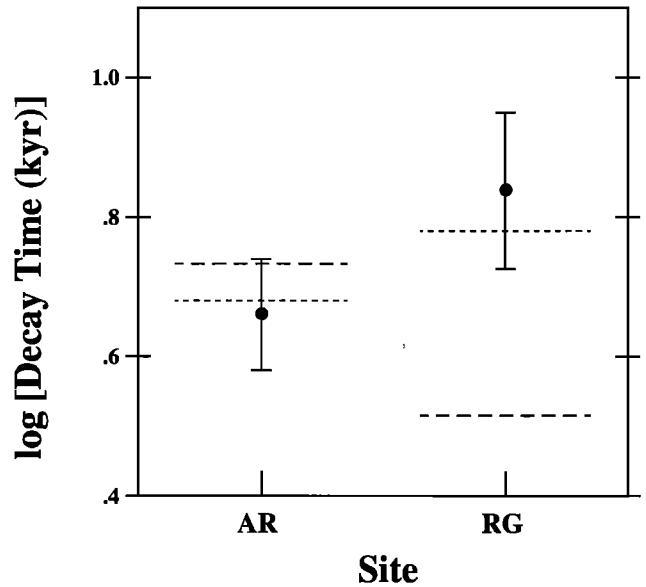


Figure 8. The prior (long dashed lines) and posterior (short dashed lines) decay times at Angerman River (AR) and Richmond Gulf (RG) associated with the inversion shown in Figure 6c. The vertical bars represent the $\pm 1\sigma$ uncertainty in the observational constraints.

the kernels for the Richmond Gulf and Angerman River sites in Figure 5. As an example, $A_{AR}^A = 20.97 \pm 0.10$ and $A_{AR}^B = 20.88 \pm 0.14$ when these averages are applied to the posterior model in Figure 6c. These are in complete agreement with the bound on the Haskell value derived from earlier inversions based on the Angerman River decay time alone. We can define analogous averages, A_{RG}^A and A_{RG}^B , as the linear combination of the posterior model with a weighting ($W(r)$ in equation (6)) based on the Frechet kernels for the Richmond Gulf decay time in Figures 5a and 5b, respectively. These averages, applied to the posterior model in Figure 6c, yield the estimates 21.32 ± 0.15 and 21.33 ± 0.19 .

We conclude that the Richmond Gulf decay time is reconciled by models in which A_{RG}^A or $A_{RG}^B \sim 21.3$. This is accomplished, in the posterior model of Figure 6c, by increasing the viscosity in the mantle between 670 and 2000 km depth. In order that the same model simultaneously satisfy the Haskell constraint (on A_{AR}), this increase in viscosity must be compensated by a reduction in the viscosity of the upper mantle from its prior value. Since 30% of the area in the averaging kernel for Richmond Gulf resides in the upper mantle, to achieve an average of A_{RG}^A or $A_{RG}^B \sim 21.3$ in the case when the upper mantle is relatively weak requires a significant increase in viscosity in the top half of the lower mantle relative to the upper mantle value (see Figure 6c). Indeed, the volumetric average of the posterior model in the upper mantle (A_{UM}) is 20.68 (or 4.8×10^{20} Pa s), while in the top 1000 km of the lower mantle it is 21.65 (or 4.5×10^{21} Pa s). This inference, which suggests a broad order of magnitude increase in the viscosity over the top ~ 1700 km of the mantle, is in agreement with the results of *Lambeck et al. [1990]*. Finally, the de-

crease in viscosity with depth in the lower mantle of the posterior model in Figure 6c is not significant. It is merely a reflection, as in earlier inversions, of a trend toward the prior value of viscosity within a region (the bottom half of the lower mantle) in which the observational data provide little constraint.

The inference appearing in Figure 6c requires two very important caveats. First, in simultaneously inverting data from Fennoscandia and Hudson Bay we have neglected the potentially significant influence of lateral heterogeneities in the mantle strength. Second, the postglacial RSL record in Hudson Bay is suspect since it is characterized by decay times which are not self-consistent [e.g., *Han and Wahr*, 1995; *Mitrovica and Peltier*, 1995]. In particular, RSL curves in northern Hudson Bay suggest shorter decay times ($\sim 2\text{--}5$ kyr) than those in the southern portions of the Bay [*Mitrovica and Peltier*, 1995]. In Figure 6d we show the results of an inversion of decay times determined from the RSL record at Angerman River as well as at Southampton Island, Ottawa Island and Ungava Peninsula in northern Hudson Bay. We have found that the inversion of these four data points is less stable than the two-site inversion of Figure 6c, and accordingly, the prior variance of the model parameters has been reduced from 4.0 to 1.0. The relatively short decay times determined from the northern Hudson Bay RSL record lead to a weaker viscosity within the top portion of the lower mantle, relative to Figure 6c, and to essentially no viscosity contrast across the 670-km interface. Indeed, the averaging (Frechet) kernels associated with decay time predictions for either Angerman River or northern Hudson Bay (the latter are similar in form to the Richmond Gulf kernels in Figure 5) yield posterior estimates near 20.9 when applied to the inverted profile in Figure 6d. Clearly, a conclusive inference of viscosity based on a simultaneous inversion of decay time data from Hudson Bay and Fennoscandia must await the determination of a self-consistent Hudson Bay RSL record.

Summary

We have revisited *Haskell's* [1935] classic inference of mantle viscosity. We find that of the data used by Haskell (RSL curves at Angerman River and Oslo) only the decay time determined from the Angerman River RSL curve provides an inference of mantle viscosity which is insensitive to variations in either the late Pleistocene ice load history or the assumed lithospheric thickness. Using forward and inverse analyses, we have derived an "average" viscosity of $\sim 0.65\text{--}1.10 \times 10^{21}$ Pa s, which is in agreement with Haskell's original results. More importantly, we show that the averaging kernel is associated with a radial region which extends to 1400 km depth (i.e., well into the lower mantle). Since the late 1970s the Haskell value has been commonly interpreted as a constraint on the average viscosity in the upper mantle alone [e.g., *Tushingham and Peltier*, 1992]. We conclude that this view is, in general, invalid. Forward and inverse analyses indicate that the

viscosity within both the upper and lower mantle can vary significantly while still satisfying the remarkably robust constraint on viscosity provided by the decay time of the postglacial RSL history at Angerman River.

In order to reduce the nonuniqueness associated with the Haskell value of viscosity, we have also considered inferences of viscosity which invoke decay time constraints at Richmond Gulf, Quebec, and in northern Hudson Bay. These decay times are sensitive to variations in mantle viscosity to a depth of about 2000 km. A simultaneous inversion of the decay times at Angerman River and Richmond Gulf suggests about an order of magnitude increase in the viscosity between the upper mantle and the top 1000 km of the lower mantle. In this regard the inference is consistent with those of *Lambeck et al.* [1990] in their analysis of the RSL database from NW Europe. The inference must, however, be considered preliminary, since the present RSL data base from Hudson Bay does not yield a consistent set of postglacial decay times. Indeed, an inversion of postglacial decay times from Angerman River and northern Hudson Bay show no significant increase in mantle viscosity down to a depth of ~ 1700 km.

The decay times from Fennoscandia and Hudson Bay do not rule out a significant increase in viscosity in the bottom half of the lower mantle. Independent constraints based on the low-degree, long-wavelength signature of GIA (e.g., rotational anomalies, secular variations in the low-degree zonal harmonics of the Earth's geopotential, etc.) may be important in this regard. However, a number of these observables, in particular the J_1 harmonics, are significantly influenced by the recent melting of small ice sheets and glaciers and any ongoing mass flux associated with the large polar ice caps [e.g., *Trupin et al.*, 1992; *Mitrovica and Peltier*, 1993c]. Furthermore, predictions of the wander of the rotation pole with respect to the surface geography are sensitive to the lithospheric thickness [e.g., *Wu and Peltier*, 1984]. Alternatively, joint inversions of GIA data and convectively supported long wavelength geoid anomalies [e.g., *Richards and Hager*, 1984] may provide constraints on mantle viscosity right down to the CMB. The first such inversions (A. M. Forte and J. X. Mitrovica, A new inference of mantle viscosity based on a joint inversion of post-glacial rebound data and long-wavelength geoid anomalies, submitted to *Geophysical Research Letters*, 1995) suggest that the insensitivity of most GIA data sets to variations in viscosity within the bottom half of the lower mantle may permit a reconciliation of the apparently contradictory inferences based on the two data types. These inversions are characterized by an average viscosity in the bottom half of the lower mantle of $\sim 10^{22}$ Pa s [see *Mitrovica and Davis*, 1995].

Decay times estimated from RSL histories at the center of previously glaciated regions such as Fennoscandia and Laurentia provide constraints on mantle viscosity of unparalleled robustness. Certainly inferences based on these constraints are superior to the usual procedure of considering raw RSL data. Unfortunately, the

existing data set from these regions is not at all extensive. Indeed, the RSL curve from Angerman River is the most commonly (and often the only) cited example from Fennoscandia. Furthermore, of the small number of RSL sites commonly considered in southern Hudson Bay (perhaps three), only one (Richmond Gulf) has a sufficient number of age-height pairs to ensure a confident estimate of the characteristic decay time of the RSL variation. A renewed effort to improve this situation is clearly warranted.

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